

ALGEBRA 2 6.4 - RATIONAL EXPONENTS

Review: When a base is raised to a rational (Fraction) exponent, that exponent represents both a power (top) and a root/index (bottom)

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} \text{ or } (\sqrt[n]{a})^m$$

↖ power
↖ root/index

I. SIMPLIFY

1. $36^{\frac{1}{2}}$ root
 change $\sqrt{36} = 6$
 calculator: $= 6$

2. $64^{\frac{1}{3}}$
 change to $\sqrt[3]{64}$
 calculator: $\boxed{4}$

3. $81^{\frac{1}{4}}$
 $\sqrt[4]{81}$
 $\boxed{3}$

4. $6^{\frac{1}{2}} \cdot 18^{\frac{1}{2}}$
 radical form:
 $\sqrt{6} \cdot \sqrt{18} = \sqrt{6 \cdot 18}$
 $\sqrt{108} = \sqrt{36 \cdot 3} = 6\sqrt{3}$
 $\boxed{6\sqrt{3}}$

5. $4^{\frac{1}{3}} \cdot 6^{\frac{1}{3}}$
 $\sqrt[3]{4 \cdot 6} = \sqrt[3]{24} = 2\sqrt[3]{3}$
 $\boxed{2\sqrt[3]{3}}$

~~$\frac{\text{power}}{\text{root/index}}$~~ on the radical.

II. REWRITE AS A RADICAL

6. $x^{\frac{1}{4}}$ power
index
 $\sqrt[4]{x^1}$
 $\boxed{\sqrt[4]{x}}$

6. $y^{\frac{2}{3}}$
 $\sqrt[3]{y^2}$
 $\boxed{\sqrt[3]{y^2}}$

7. $z^{-\frac{4}{5}}$
 $\frac{1}{z^{\frac{4}{5}}} = \frac{1}{\sqrt[5]{z^4}}$
 $\boxed{\frac{1}{\sqrt[5]{z^4}}}$

$$\text{bottom} \rightarrow \sqrt[n]{a^{\text{top}}} = a^{m/n}$$

III. REWRITE IN EXPONENTIAL FORM

8. $\sqrt[3]{x^4}$

$$x^{4/3}$$

9. $\sqrt{a^5}$

$$a^{5/2}$$

10. $\sqrt[3]{(5x^2)^4}$

$$(5x^2)^{4/3}$$

11. $\sqrt[3]{(5x)^2}$

$$(5x)^{2/3}$$

12. $(\sqrt[3]{5x})^2$

$$(5x)^{2/3}$$

IV. EXPRESS AS A POWER OF A SINGLE BASE: Change to exponential form then add the exponents.

13. $\sqrt{2} \cdot \sqrt[3]{2}$

$$2^{1/2} \cdot 2^{1/3} = 2^{3/6} \cdot 2^{2/6} = 2^{5/6}$$

14. $\sqrt[4]{7} \cdot \sqrt[3]{7}$

$$7^{1/4} \cdot 7^{1/3} = 7^{3/12} \cdot 7^{4/12} = 7^{7/12}$$

15. $\frac{\sqrt{y^3}}{\sqrt[4]{y^3}}$

$$= \frac{y^{3/2}}{y^{3/4}} = y^{3/4}$$

16. $\sqrt[5]{4} \cdot \sqrt[3]{32} \cdot \sqrt{2^5}$

$$2^{2/5} \cdot 2^{10/3} \cdot 2^{5/2} = 2^{4/10} \cdot 2^{20/6} \cdot 2^{25/6} = 2^{4/10} \cdot 2^{25/6} = 2^{29/10}$$

17. $\frac{\sqrt{9}}{\sqrt[4]{27}}$

$$\frac{\sqrt{3^2}}{\sqrt[4]{3^3}} = \frac{3^{2/2}}{3^{3/4}} = \frac{3^{4/4}}{3^{3/4}} = 3^{1/4}$$

V. EVALUATE THE RATIONAL EXPONENTS.

18. $9^{3/2}$

$$27$$

19. $27^{4/3}$

$$81$$

$$(3\sqrt{27})^4$$

20. $4^{-3/2}$

$$\frac{1}{4^{3/2}} = \frac{1}{8}$$